Course 02323 Introduction to Statistics & 02402 Statistics (Polytechnical Foundation)

Lecture 3: Random variables and continuous distributions

DTU Compute
Technical University of Denmark
2800 Lyngby – Denmark

Overview

1. Summary of Previous Week
2. Continuous Distributions
   - Density and Distribution Functions
   - Mean, variance, and covariance
3. Specific continuous distributions
   - The Uniform distribution
   - The Normal distribution
   - The log-normal distribution
   - The Exponential distribution
4. Rules for stochastic variables
5. Special distributions used later in the course
   - χ^2-distribution
   - t-distribution
   - F-fordeling

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Summary: Week 2, Discrete Distributions

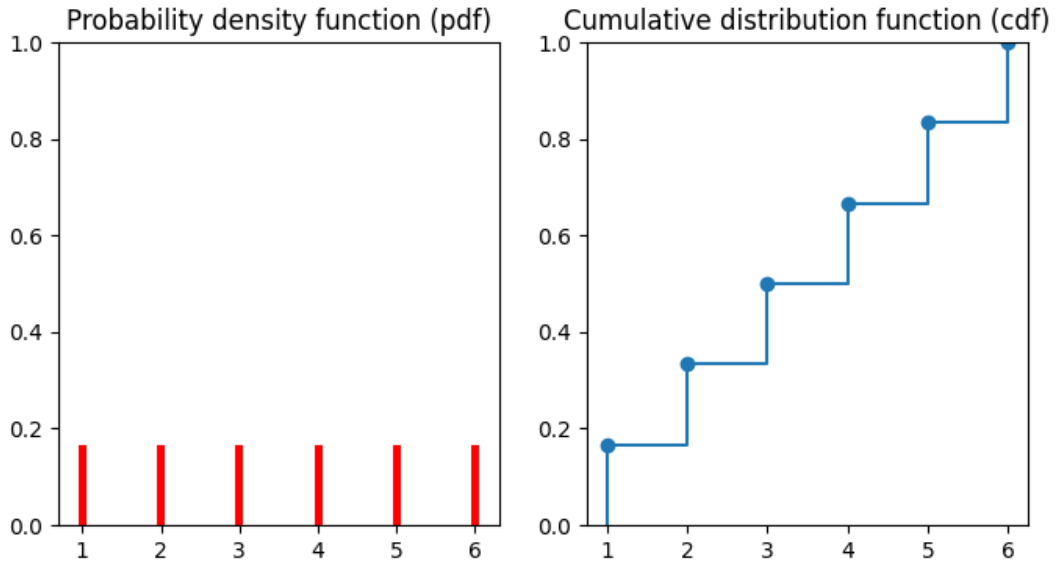
A stochastic variable: X

Probability density function (*pdf* / *pmf*):

$$f(x) = \mathrm{P}(X = x)$$

Cumulative distribution functior (*cdf*):

$$F(x) = \mathrm{P}(X \leq x)$$



Expected value:

$$\mathrm{E}[X] = \sum_{x \in A} x \, \mathrm{P}(X = x) = \mu$$

Variance:

$$\mathrm{V}[X] = \sum_{x \in A} (x - \mu)^2 \, \mathrm{P}(X = x) = \sigma^2$$

Alternatively:

$$\mathrm{V}[X] = \mathrm{E}[(X - \mu)^2]$$

Exam with discrete distribution

Exam from May 2021, questions 19 and 20

https://02402.compute.dtu.dk/previousexams/2021may_02402_solution_en.pdf

Exam question

Question VII.1 (19)

What is the probability of exactly three costumers arriving in a randomly selected 1-minute period?

1* ☐ **stats.poisson.pmf(k=3, mu=3) = 0.2240**

2 ☐ **stats.poisson.cdf(k=2, mu=3) = 0.4232**

3 ☐ **1 - stats.poisson.cdf(k=2, mu=3) = 0.5768**

4 ☐ **stats.poisson.cdf(k=3, mu=3) = 0.6472**

5 ☐ **1 - stats.poisson.pmf(k=3, mu=3) = 0.7760**

Exam question

Question VII.2 (20)

What is the probability of at least three arrivals in a randomly selected 1-minute period?

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Continuous Random Variables

Random variable X.

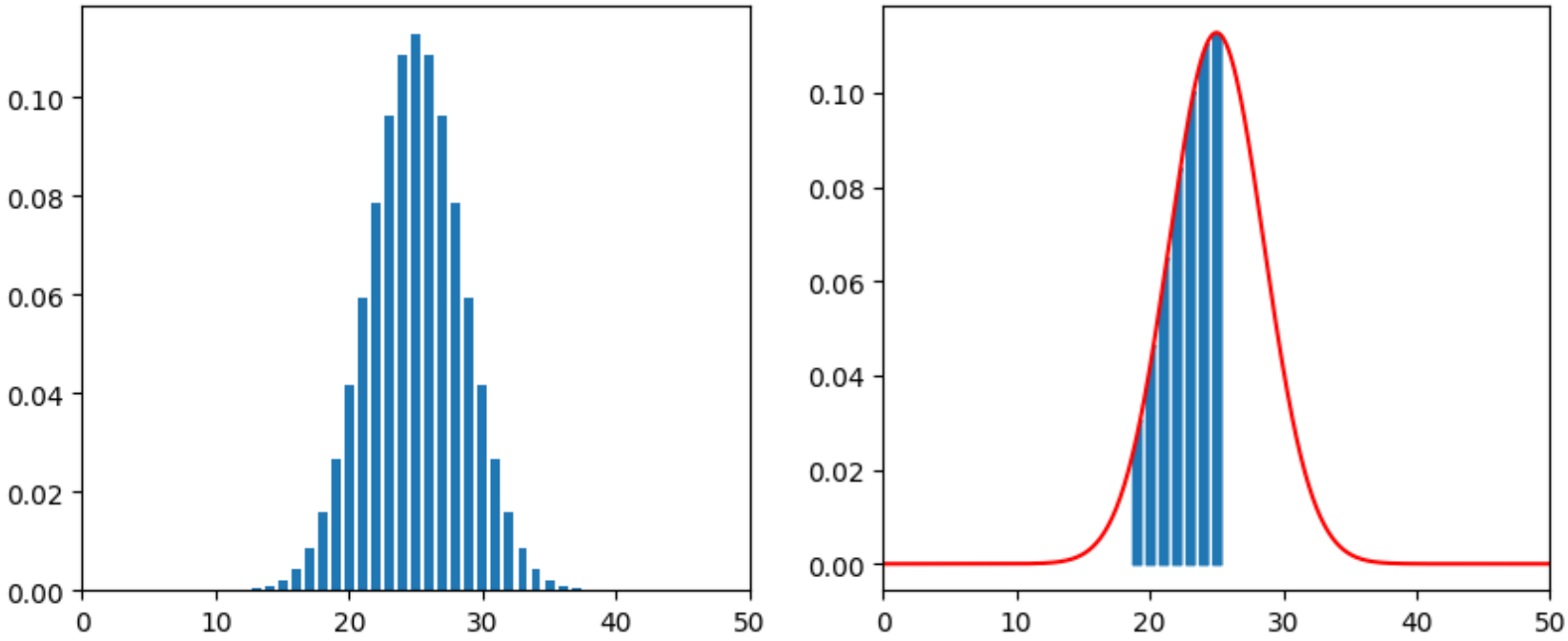
The sample space S is now continuous.

Examples:

- Height of students
- Measurement of wind speed
- Time to cycle to DTU
- Measurement of blood sugar in patients
- ...

Continous probability distribution

A discrete probability distribution (left) assigns probabilities to distinct, separate values shown as bars, whereas a continuous probability distribution (right) assigns probability over an interval of values represented by a smooth curve where probabilities correspond to areas under the curve.



The density function, Definition 2.32

- The probability density function (pdf) for a random variable is denoted by $f(x)$.

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$$P(a < X \leq b) = \int_a^b f(x)\,dx\,.$$

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- *No* direct probability for pdf. In fact, $P(X = x) = 0$ for all x.
- The density function $f(x)$ for the distribution of a continuous random variable satisfies that

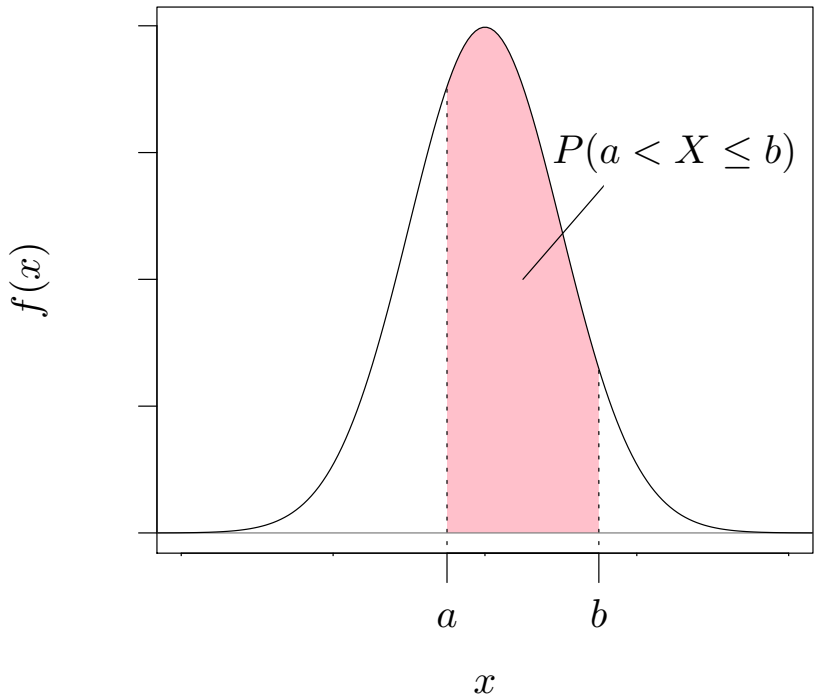
$$f(x) \geq 0 \text{ for all } x \quad \text{and} \quad \int_{-\infty}^{\infty} f(x)\,dx = 1\,.$$

The density function (Continuous)

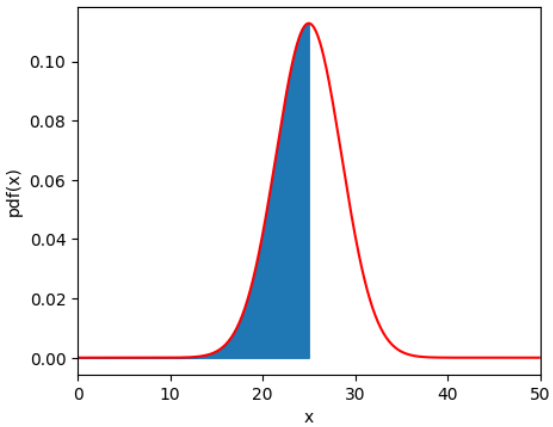
The probability of observing the outcome of X in the range between a and b is the area below the probability density function (pdf) over that interval, i.e.,

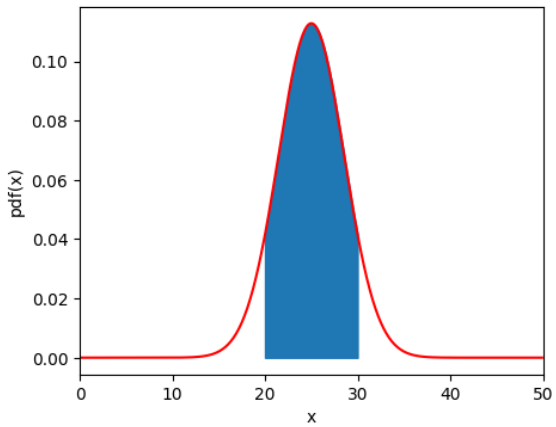
$$P(a \leq X \leq b) = \int_a^b f_X(x)\,dx,$$

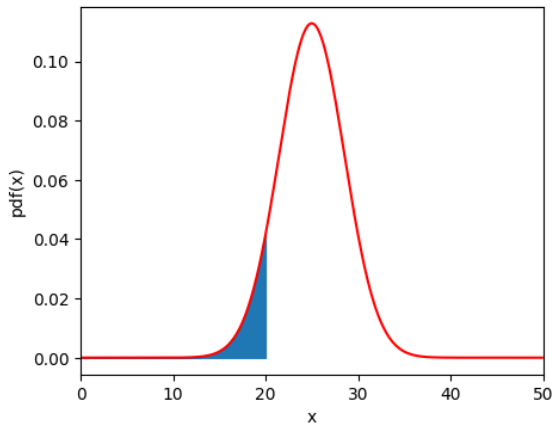
as illustrated by the coloured area under the curve.



Kahoot





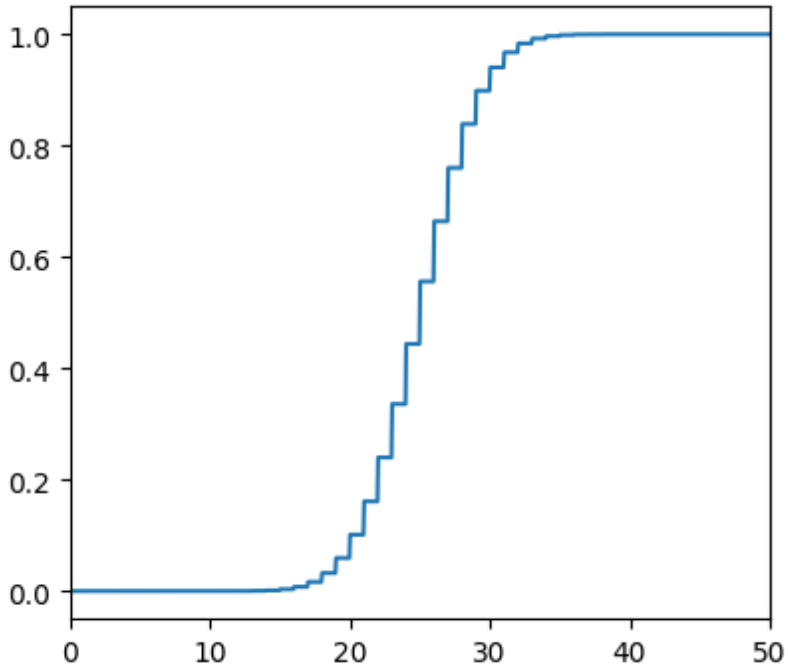


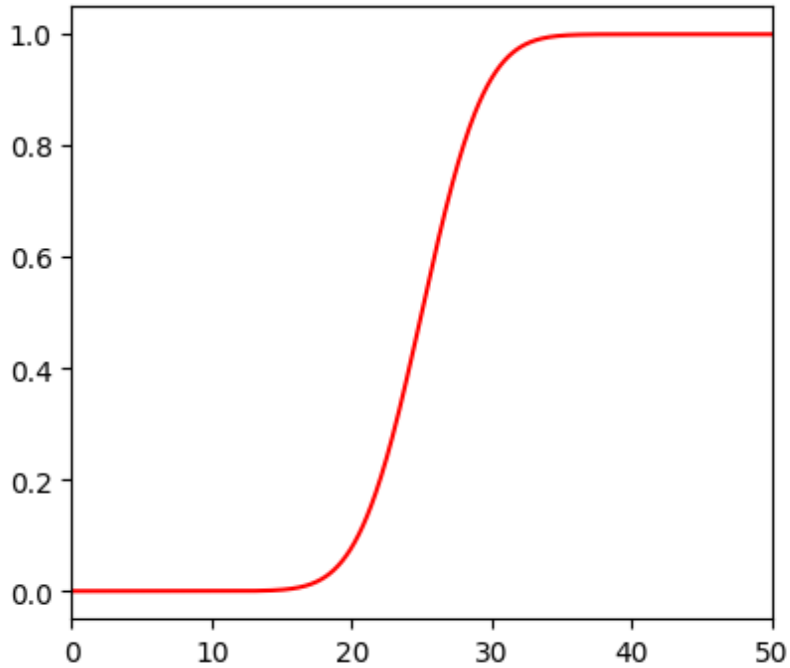
Kahoot!

(x4)

Distribution Function for Continuous Variables

The cumulative distribution function (CDF) of a discrete probability distribution increases in steps at the possible outcome values, whereas the CDF of a continuous probability distribution is a smooth, continuous curve without jumps.





The distribution function, Definition 2.33

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$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\, dt\,.$$

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- Note that as a consequence of this definition,

$$f(x) = F'(x)\,.$$

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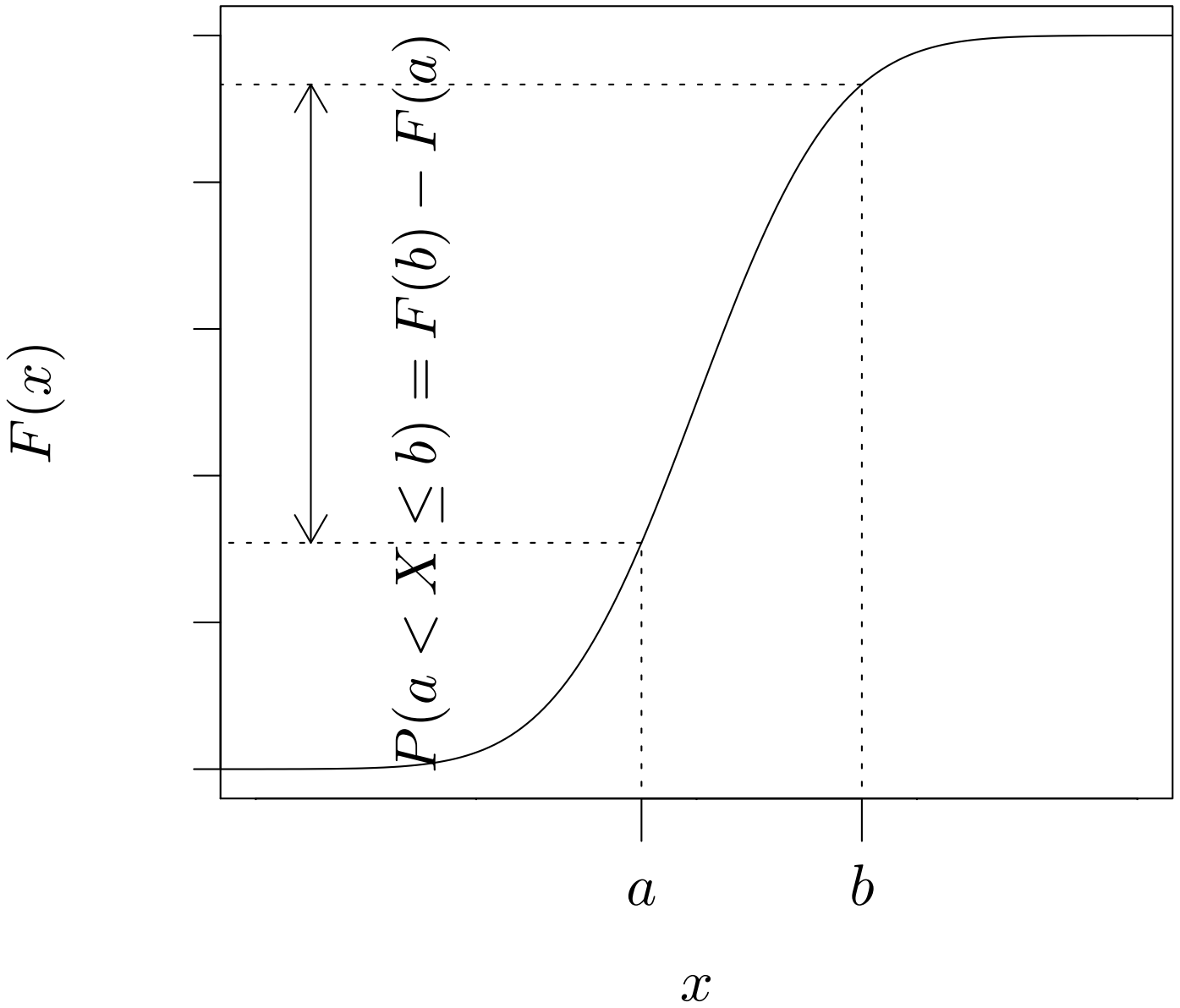
- Note that as a consequence of this definition,

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- It's particularly useful to note that

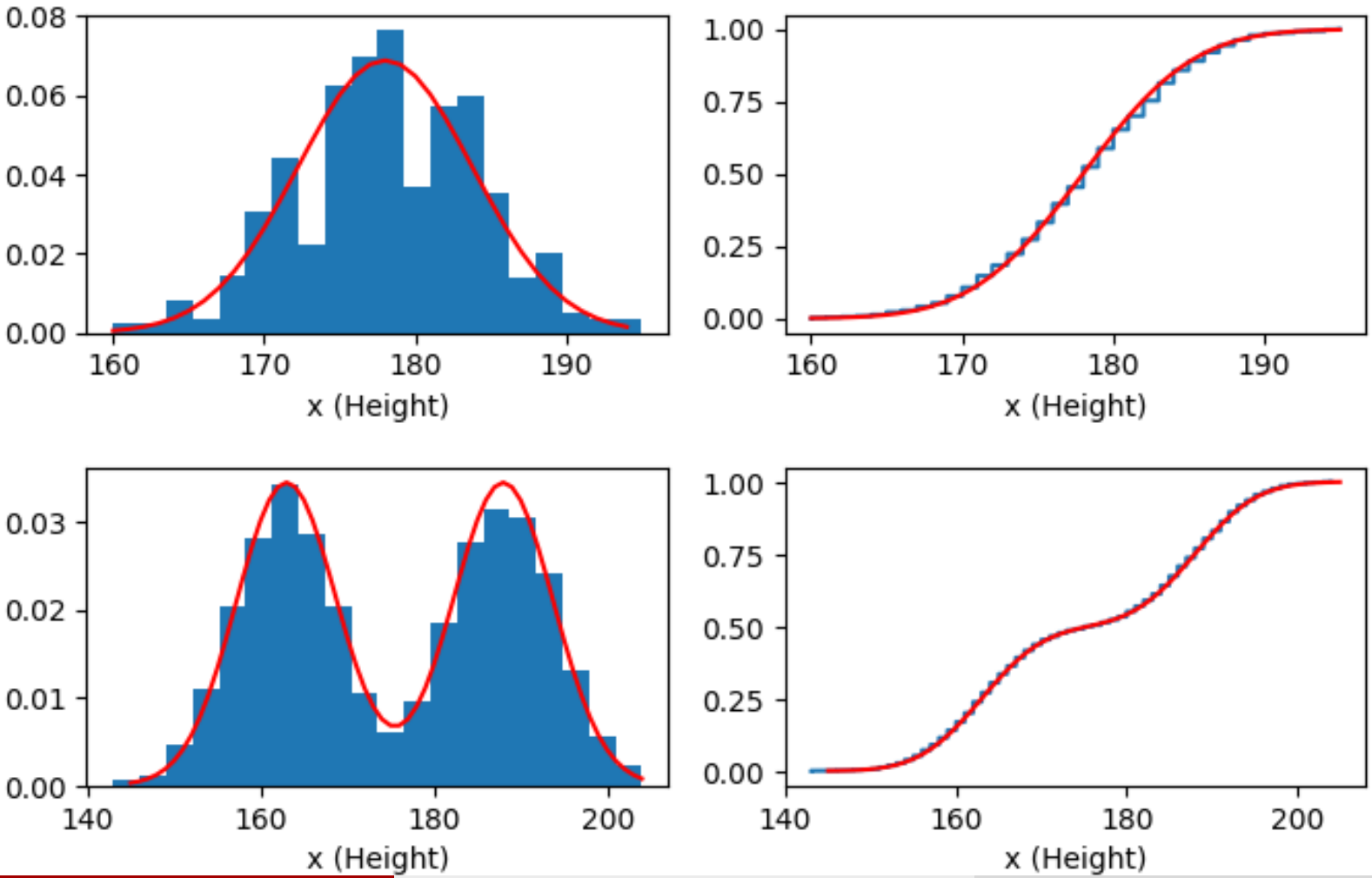
$$P(a < X \leq b) = \int_{a}^{b} f(x)\,dx \; = F(b) - F(a).$$

Continuous distribution function

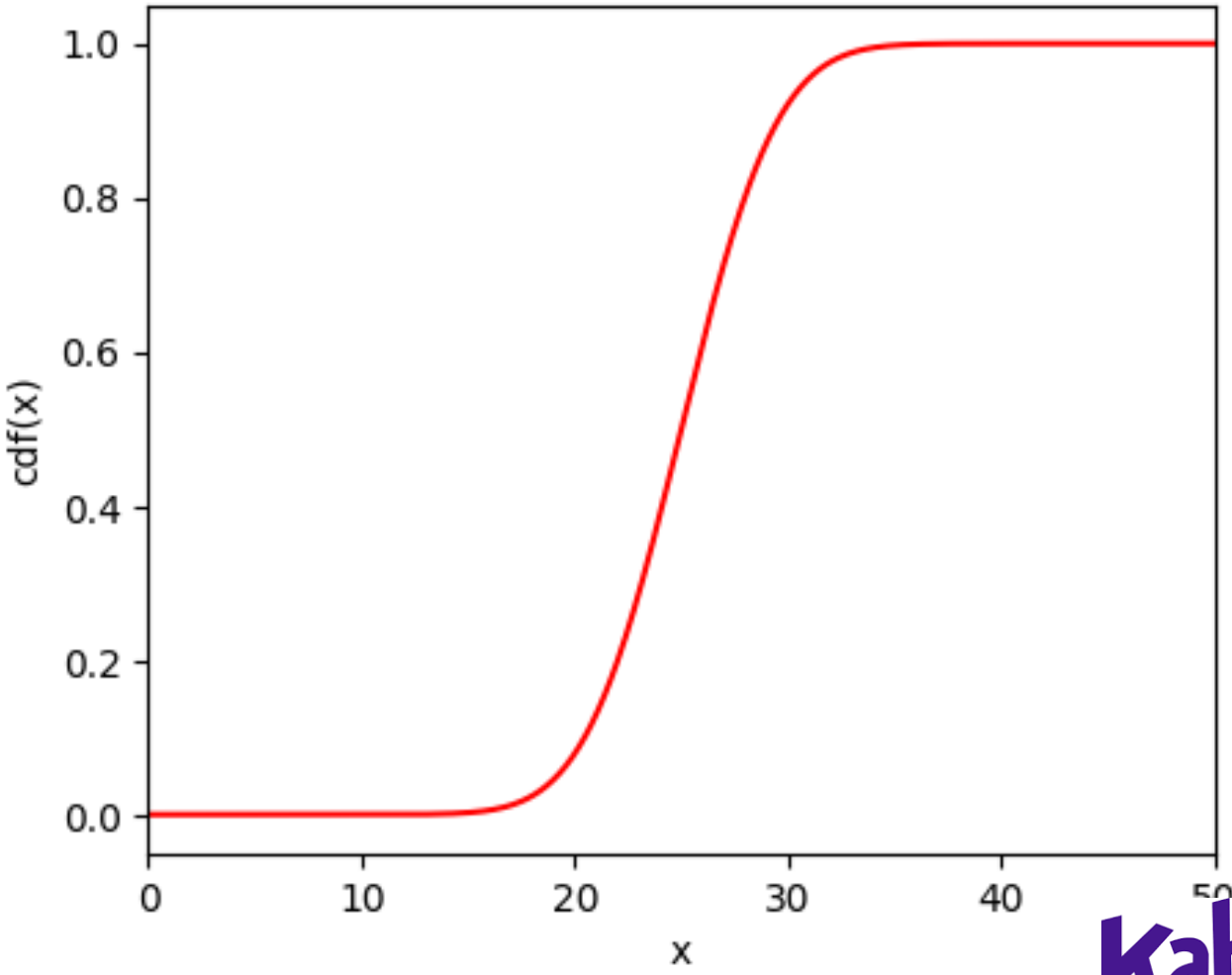


Examples: Continuous density and distribution curves

We observe that a **unimodal density** (top row) produces a smooth **S-shaped CDF** with the steepest increase around the single peak, whereas a **bimodal density** (bottom row) leads to a CDF of two clusters of observations in the distribution.



Kahoot



Kahoot!

(x2)

Mean, continuous random variable, Definition 2.34

The mean/expected value of a continuous random variable:

$$\mu = \mathrm{E}[X] = \int_{-\infty}^{\infty} x\, f(x)\, dx$$

Mean, continuous random variable, Definition 2.34

The mean/expected value of a continuous random variable:

$$\mu = \mathrm{E}[X] = \int_{-\infty}^{\infty} x f(x)\, dx$$

Compare with the mean of a discrete random variable:

$$\mu = \mathrm{E}[X] = \sum_{\text{all } x} x f(x)$$

Variance, continuous random variable, Definition 2.34

The variance of a continuous random variable:

$$\sigma^2 = \mathrm{E}[(X-\mu)^2] = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\,dx$$

Variance, continuous random variable, Definition 2.34

The variance of a continuous random variable:

$$\sigma^2 = \mathrm{E}[(X-\mu)^2] = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\, dx$$

Compare with the variance of a discrete random variable:

$$\sigma^2 = \mathrm{E}[(X-\mu)^2] = \sum_{\text{all } x} (x-\mu)^2 f(x)$$

Example Exam 02323 December 2025

Exercise II

The discrete random variable X has the following distribution:

x	1	3	5	6
$f(x)$	0.1	0.4	0.3	0.2

where $f(x) = P(X = x)$ is the probability density function (pdf).

Question II.1 (2)

What is the mean $\mathbf{E}[X]$ and variance $\mathbf{V}[X]$?

1 ☐ $\mathbf{E}[X] = 3.75$ and $\mathbf{V}[X] = 1.92$

2 ☐ $\mathbf{E}[X] = 4.00$ and $\mathbf{V}[X] = 1.92$

3 ☐ $\mathbf{E}[X] = 4.40$ and $\mathbf{V}[X] = 2.40$

4 ☐ $\mathbf{E}[X] = 3.75$ and $\mathbf{V}[X] = 2.40$

5 ☐ $\mathbf{E}[X] = 4.00$ and $\mathbf{V}[X] = 2.40$

6 ☐ Don't know / No answer

Example Exam 02323 December 2025

$$
\begin{aligned}
E[X] &= \sum x \cdot f(x) \\
&= 1 \cdot 0.1 + 3 \cdot 0.4 + 5 \cdot 0.3 + 6 \cdot 0.2 \\
&= 0.1 + 1.2 + 1.5 + 1.2 \\
&= 4.0
\end{aligned}
$$

$$
\begin{aligned}
V[X] &= \sum (x - E[X])^2 \cdot f(x) \\
&= (1-4)^2 \cdot 0.1 + (3-4)^2 \cdot 0.4 + (5-4)^2 \cdot 0.3 + (6-4)^2 \cdot 0.2 \\
&= (-3)^2 \cdot 0.1 + (-1)^2 \cdot 0.4 + 1^2 \cdot 0.3 + 2^2 \cdot 0.2 \\
&= 9 \cdot 0.1 + 1 \cdot 0.4 + 1 \cdot 0.3 + 4 \cdot 0.2 \\
&= 0.9 + 0.4 + 0.3 + 0.8 \\
&= 2.4
\end{aligned}
$$

Covariance, Definition 2.58

The covariance between two random variables:

Let X and Y be two random variables. Then, the covariance between X and Y is

$$\text{Cov}(X,Y) = E[(X - E[X])(Y - E[Y])]$$

Relationship between covariance and independence:

If two random variables are *independent* their covariance is 0. *The reverse is not necessarily true!*

Example, Exam 02323 May 2026

Exercise III

Let X and Y be independent random variables. The expected values are $\mathbb{E}[X] = 5$ and $\mathbb{E}[Y] = -3$, while the variances are $\mathbb{V}[X] = 9$ and $\mathbb{V}[Y] = 16$.

Question III.1 (6)

What is the covariance between X and Y?

1 ☐ -25

2 ☐ -5

3 ☐ 0

4 ☐ 5

5 ☐ 25

6 ☐ Don't know / No answer

Example, Exam 02323 May 2026

Question III.1 (6)

What is the covariance between X and Y?

1 ☐ -25

2 ☐ -5

3* ☐ 0

4 ☐ 5

5 ☐ 25

6 ☐ Don't know / No answer

Since the random variables are independent, the covariance between the variables must be zero.

Correlation

Definition 2.62 Correlation

Let X and Y be two random variables with $V(X) = \sigma_x^2$, $V(Y) = \sigma_y^2$, and $Cov(X, Y) = \sigma_{xy}$, then the correlation between X and Y is

$$\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y}. \tag{2-78}$$

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Specific continuous distributions

A number of statistical distributions exist (both continuous and discrete) that can be used to describe and analyze different types of problems.

Today, we'll take a closer look at the following continuous distributions:

- The uniform distribution
- The normal distribution
- The log-normal distribution
- The exponential distribution

As we did with the discrete distributions, will we use `Scipy.stats` for the continuous distributions (see documentation online).

General 'methods' for different distributions are:

`scipy.stats`	`.uniform/.norm/.lognorm/.expon`
`.rvs`	'random variates' (simulate random numbers)
`.pdf`	'probability density function' (pdf/density function)
`.cdf`	'cumulative distribution function' (distribution function)
`.ppf`	'percent point function' (inverse cdf / quantile function)
`.mean /.var /.std`	'mean'/'variance'/'standard deviation'

See also Appendix A.2.1 in the book.

Example, Exam 02323 May 2026

Question I.2 (2)

Which function in the `scipy.stats`-package outputs random numbers?

1 ☐ The `cdf`-function

2 ☐ The `pdf`-function

3 ☐ The `pmf`-function

4 ☐ The `ppf`-function

5 ☐ The `rvs`-function

6 ☐ Don't know / No answer

Example, Exam 02323 May 2026

Question I.2 (2)

Which function in the `scipy.stats`-package outputs random numbers?

1 ☐ The `cdf`-function

2 ☐ The `pdf`-function

3 ☐ The `pmf`-function

4 ☐ The `ppf`-function

5* ☐ The `rvs`-function

6 ☐ Don't know / No answer

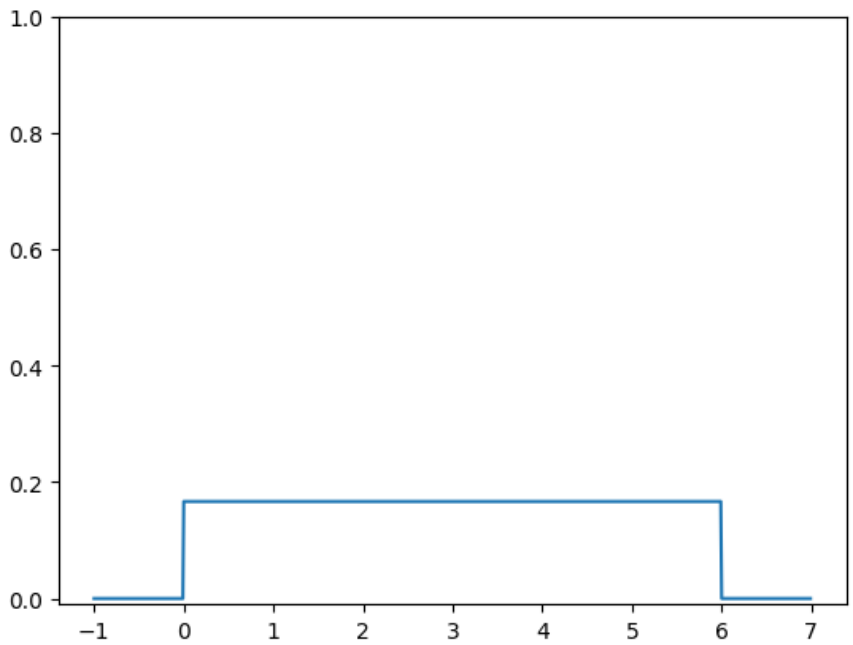
The rvs function is commonly used in Python's scipy.stats module to generate random variates from a specified probability distribution

The Uniform Distribution

The Uniform ditribution is used to describe a continuous stochastic variable X which only takes in the interval $[\alpha;\beta]$. The distribution is symmetrical and "flat".

All values in the interval are "*equally likely*"

$$X \sim U(\alpha, \beta)$$



The Uniform Distribution Density Function

Definition 2.35 Uniform distribution

Let X be a uniform distributed random variable

$$X \sim U(\alpha, \beta), \tag{2-46}$$

where α and β defines the range of possible outcomes. It has the *pdf*

$$f(x) = \begin{cases} \frac{1}{\beta-\alpha} & \text{for } x \in [\alpha, \beta] \\ 0 & \text{otherwise} \end{cases}. \tag{2-47}$$

The uniform *cdf* is

$$F(x) = \begin{cases} 0 & \text{for } x < \alpha \\ \frac{x-\alpha}{\beta-\alpha} & \text{for } x \in [\alpha, \beta) \\ 1 & \text{for } x \geq \beta \end{cases}. \tag{2-48}$$

The mean and variance of the Uniform distribution

Theorem 2.36 Mean and variance of the uniform distribution

The mean of a uniform distributed random variable X is

$$\mu = \frac{1}{2}(\alpha + \beta), \tag{2-49}$$

and the variance is

$$\sigma^2 = \frac{1}{12}(\beta - \alpha)^2. \tag{2-50}$$

Example 1

Students in a statistics course arrive at a lecture between 8:00 and 8:30. It is assumed that the arrival time can be described by a uniform distribution.

Let $X \sim U(0, 30)$ represent the "arrival time" for a randomly selected student.

Question:

What is the probability that a randomly selected student arrives between 8:20 and 8:30?

Question:

What is the probability that a randomly selected student arrives after 8:30?

Kahoot! (x2)

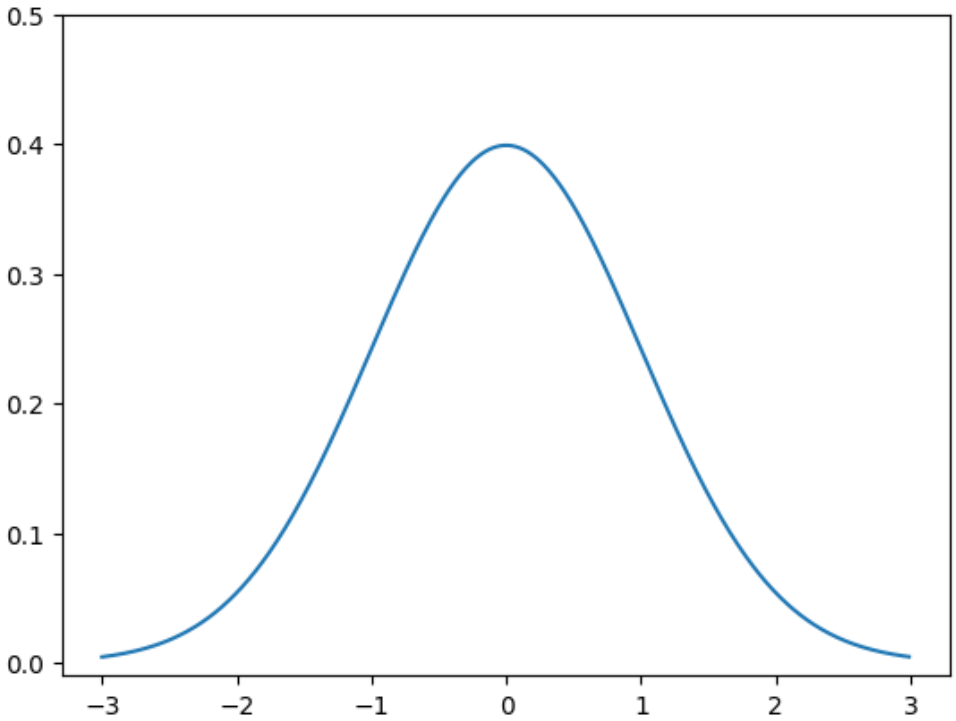
Visual Studio Code

Normal distribution

The normal distribution is used to describe a continuous random variable X that follows a characteristic bell-shaped distribution. The distribution is symmetric around the mean μ and has spread σ.

Values close to the mean are most likely

$$X \sim N(\mu, \sigma^2)$$



Normal Probability Distribution: Density & Distribution Function

Definition 2.37 Normal distribution

Let X be a normal distributed random variable

$$X \sim N(\mu, \sigma^2), \tag{2-51}$$

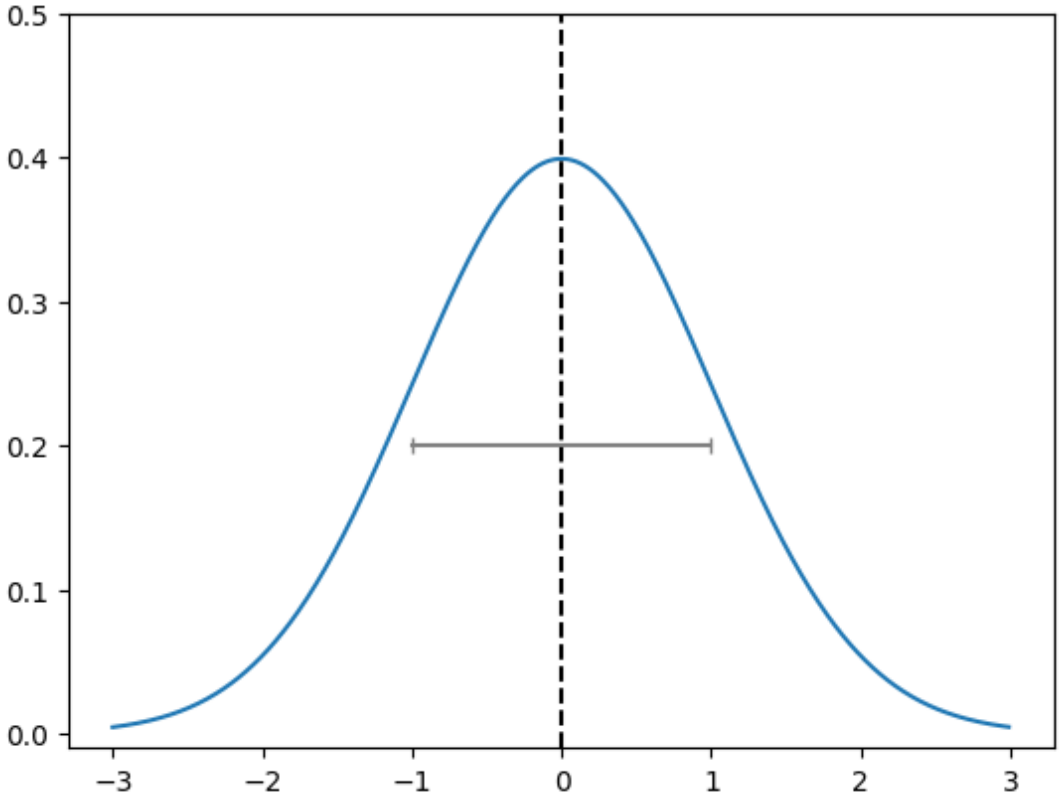
where μ is the mean and σ^2 is the variance (remember that the standard deviation is σ). Note that the two parameters are actually the mean and variance of X.
It follows the normal *pdf*

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \tag{2-52}$$

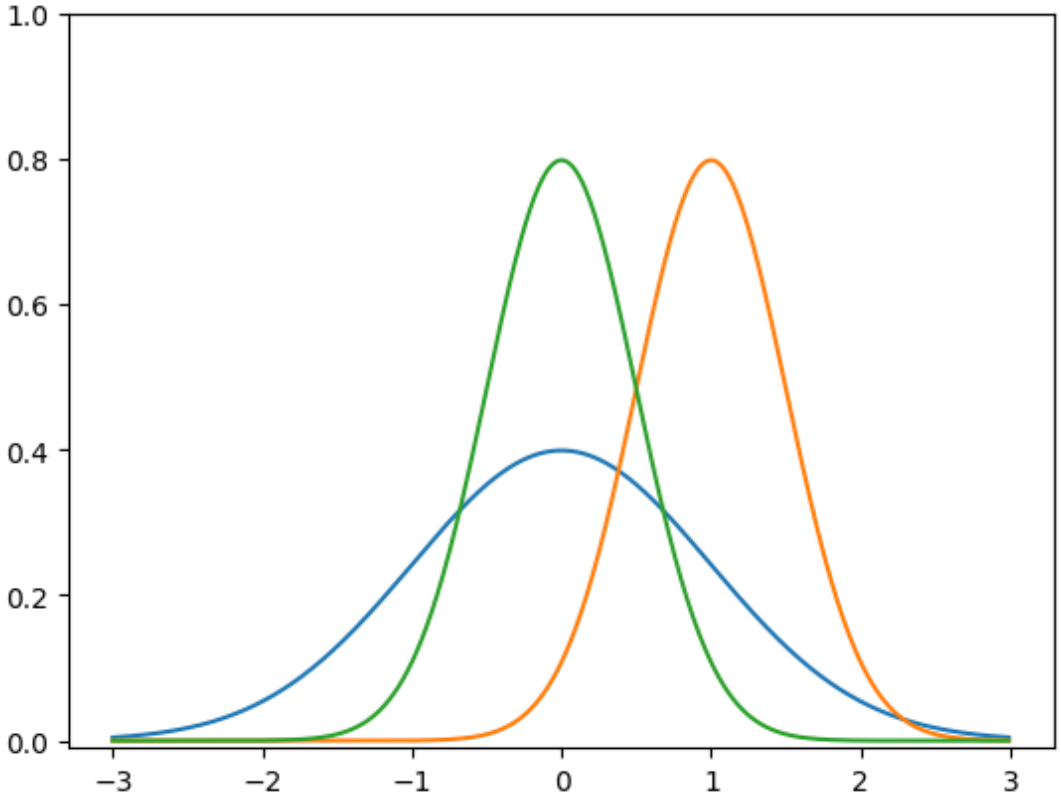
and the normal *cdf*

$$F(x) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{(u-\mu)^2}{2\sigma^2}} du. \tag{2-53}$$

Density for a normal distribution (example)



Examples of Normal Distribution



The standard normal distribution

The standard normal distribution:

$$Z \sim N(0, 1^2)$$

The normal distribution with mean 0 and variance 1.

The standard normal distribution

The standard normal distribution:

$$Z \sim N(0, 1^2)$$

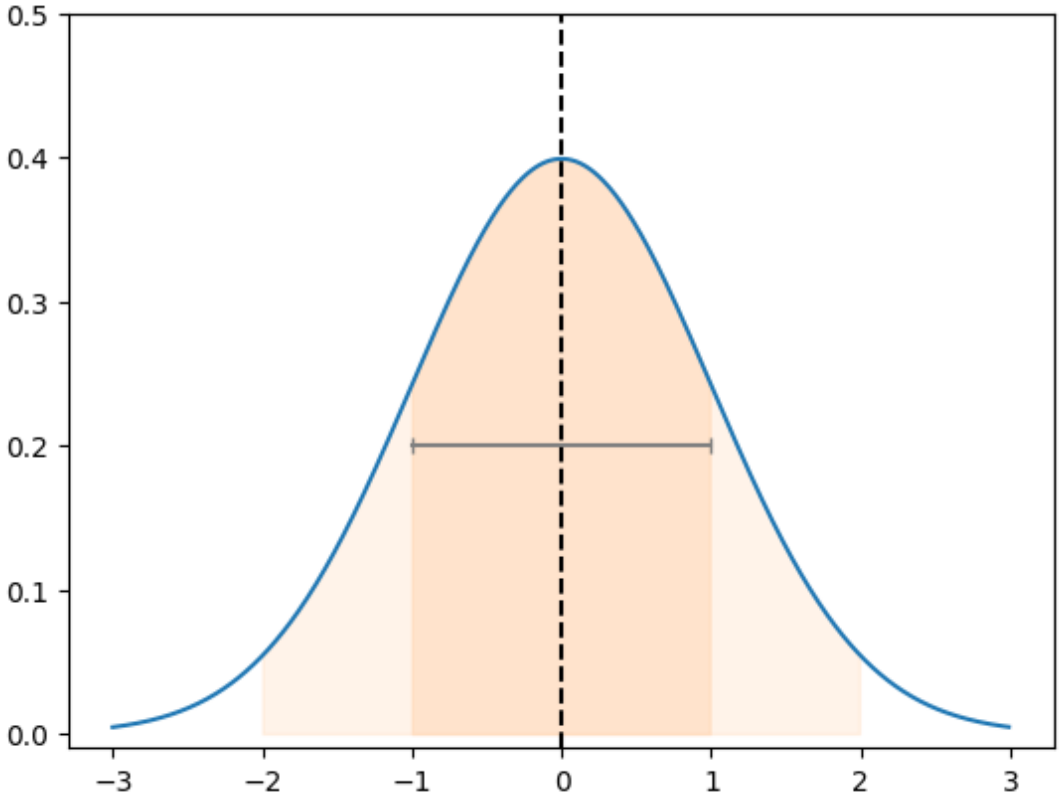
The normal distribution with mean 0 and variance 1.

Standardization:

An arbitrary normal distributed variable $X \sim N(\mu, \sigma^2)$ can be *standardized* by

$$Z = \frac{X - \mu}{\sigma}$$

The Standard Normal Distribution



Within $\mu \pm \sigma$, about 68.3% of the probability mass lies
Within $\mu \pm 2\sigma$, about 95.4% of the probability mass lies

Example 2

Measurement Error:

A given scale has a measurement error (measured in grams), Z, which can be described by a standard normal distribution, $Z \sim N(0, 1^2)$.

This means that the average measurement error is $\mu = 0$ grams and the standard deviation is $\sigma = 1$ gram.

Question:

a) What is the probability that the scale gives a result that is at least 2 grams less than the true weight of the product?
b) What is the probability that the scale gives a result that is at least 2 grams more than the true weight of the product?
c) What is the probability that the scale has a deviation of at most ± 1 gram?



Example 3

Income Distribution:

It is assumed that the salary of primary school teachers can be described by a normal distribution with a mean value of $\mu = 290$ (in 1000 DKK) and a standard deviation of $\sigma = 4$ (1000 DKK).

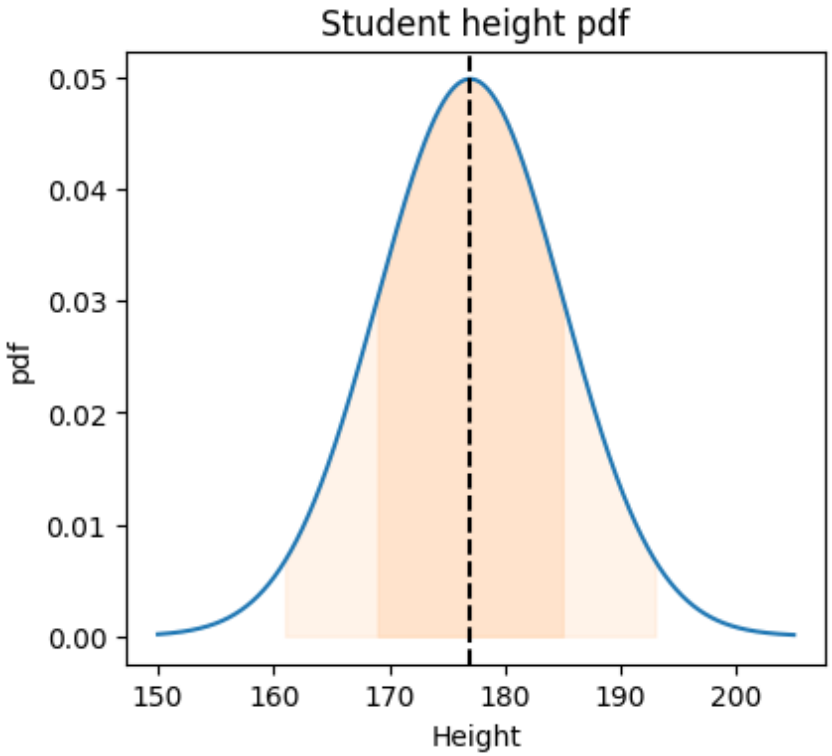
Question:

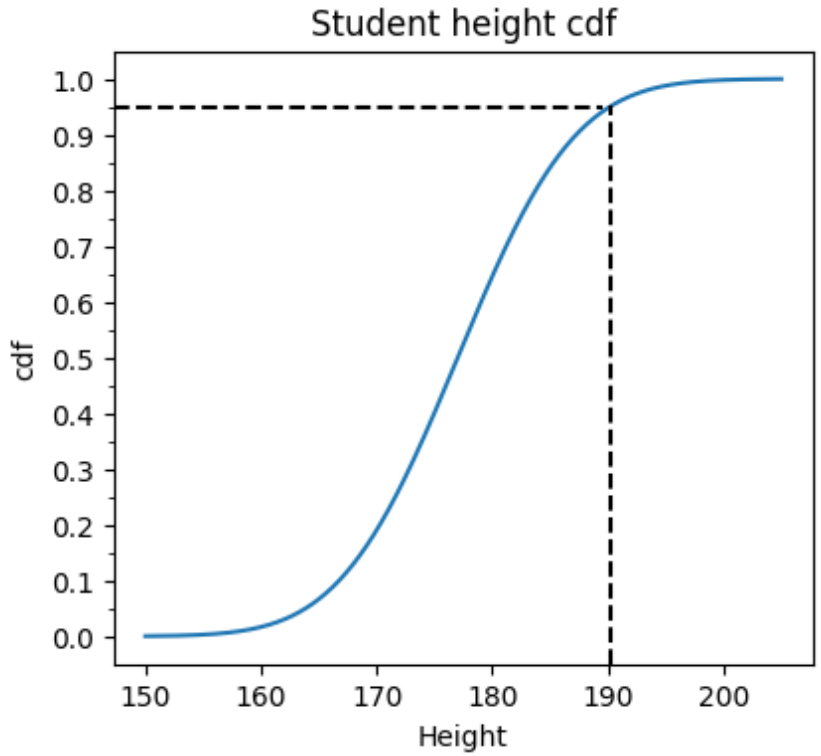
a) What is the probability that a randomly selected teacher earns more than 300,000 DKK?
b) (Inverse question) Specify a salary range (which is symmetric around the mean) that covers 95% of teachers' salaries.



Visual Studio Code

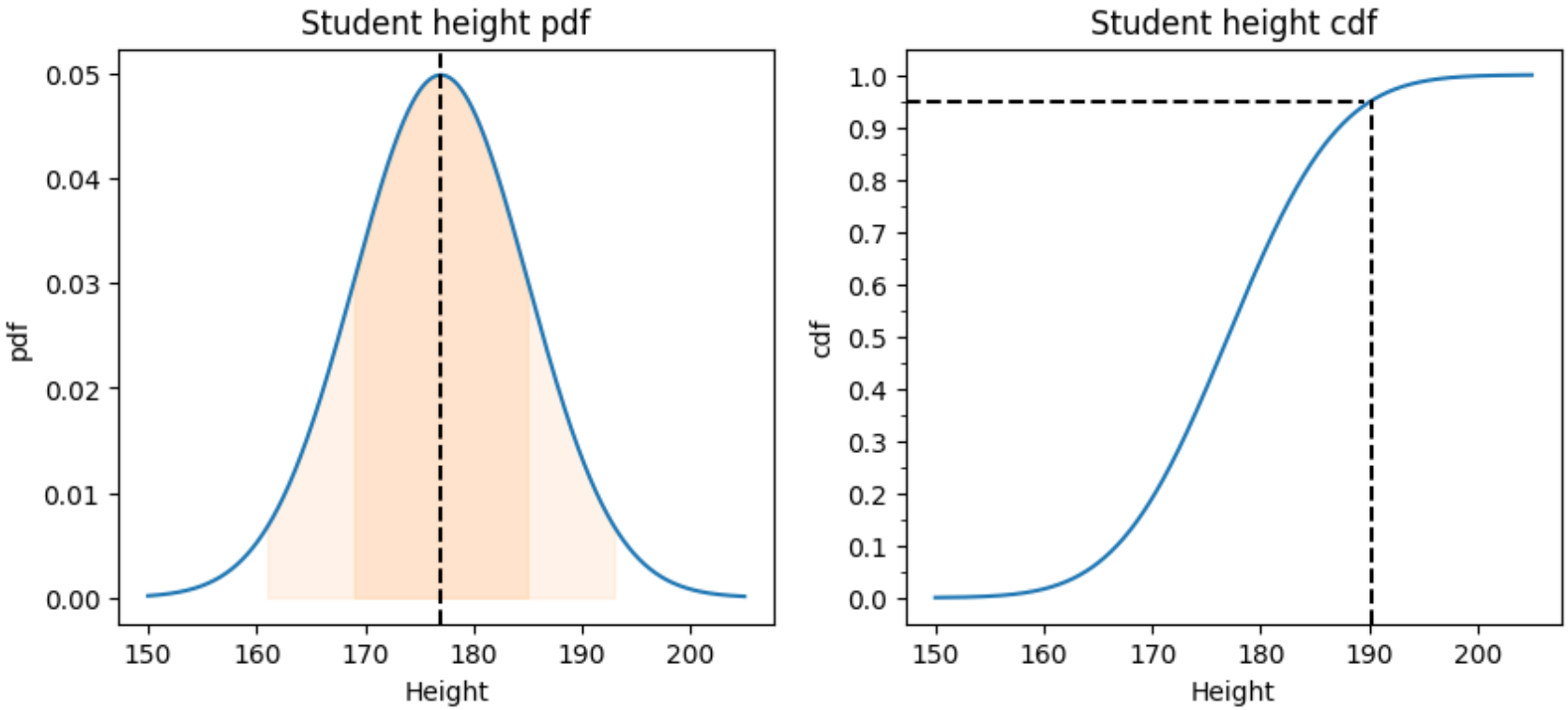
Connection between distribution and quantiles





- What is the median height? (and how do you read it from the two plots?)
- What are Q1 (the first quartile) and Q3 (the third quartile)?
- How tall should a door be if 95% of students should pass through without bending?

Connection between distribution and percentiles

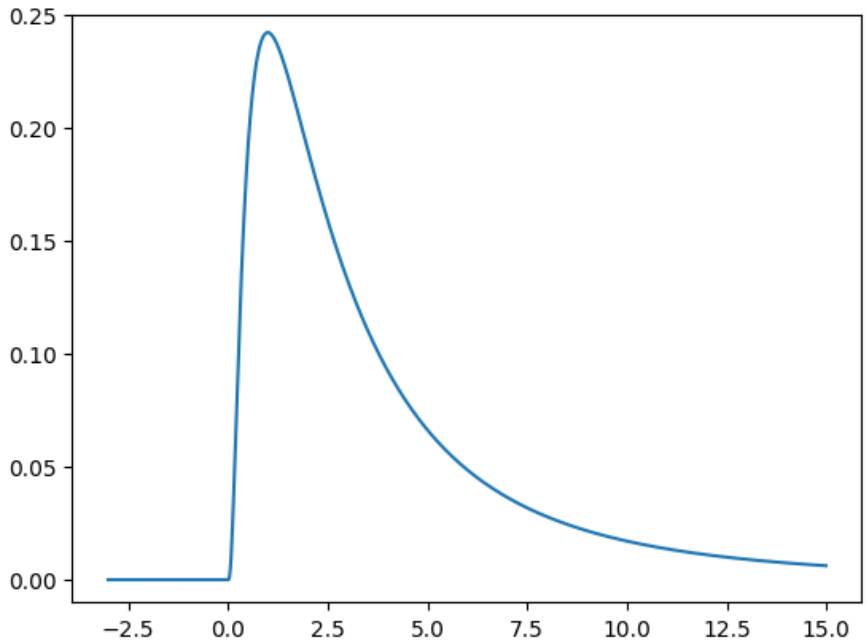


- Quantiles (percentiles)3 = "Averaged inverted cdf"
- In Python, you use ".ppf" - e.g. `stats.norm.ppf(q=0.95, loc=`μ`, scale=`σ`)`

The log-normal distribution

The lognormal distribution is used to describe a continuous stochastic variable X which follows a characteristically skewed distribution.

$$X \sim LN(\alpha, \beta^2)$$



The Log-normal distribution, Def. 2.46 & Thm. 2.47

Notation:

$X \sim LN(\alpha, \beta^2)$ (where $\beta > 0$)

The Log-normal distribution, Def. 2.46 & Thm. 2.47

Notation:

$X \sim LN(\alpha, \beta^2)$ (where $\beta > 0$)

Density function:

$$f(x) = \begin{cases} \frac{1}{\beta\sqrt{2\pi}} x^{-1} e^{-(\ln(x)-\alpha)^2/2\beta^2} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

The Log-normal distribution, Def. 2.46 & Thm. 2.47

Notation:

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$$f(x) = \begin{cases} \frac{1}{\beta\sqrt{2\pi}} x^{-1} e^{-(\ln(x)-\alpha)^2/2\beta^2} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

Mean:

$$\mu = e^{\alpha+\beta^2/2}$$

The Log-normal distribution, Def. 2.46 & Thm. 2.47

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$X \sim LN(\alpha, \beta^2)$ (where $\beta > 0$)

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$$f(x) = \begin{cases} \frac{1}{\beta\sqrt{2\pi}} x^{-1} e^{-(\ln(x)-\alpha)^2/2\beta^2} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

Mean:

$$\mu = e^{\alpha+\beta^2/2}$$

Variance:

$$\sigma^2 = e^{2\alpha+\beta^2}(e^{\beta^2} - 1)$$

The log-normal distribution

Log-normal and normal distribution:

A log-normally distributed variable $Y \sim LN(\alpha, \beta^2)$ can be transformed into a normally distributed variable X by:

$$X = \ln(Y).$$

Here X is normally distributed with mean α and variance β^2, i.e. $X \sim N(\alpha, \beta^2)$.

The log-normal distribution

Log-normal and normal distribution:

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By standardizing X through

$$Z = \frac{X - \mathrm{E}[X]}{\sqrt{\mathrm{V}[X]}} = \frac{\ln(Y) - \alpha}{\beta}$$

you get a standard normally distributed variable $Z \sim N(0, 1)$.

The log-normal distribution

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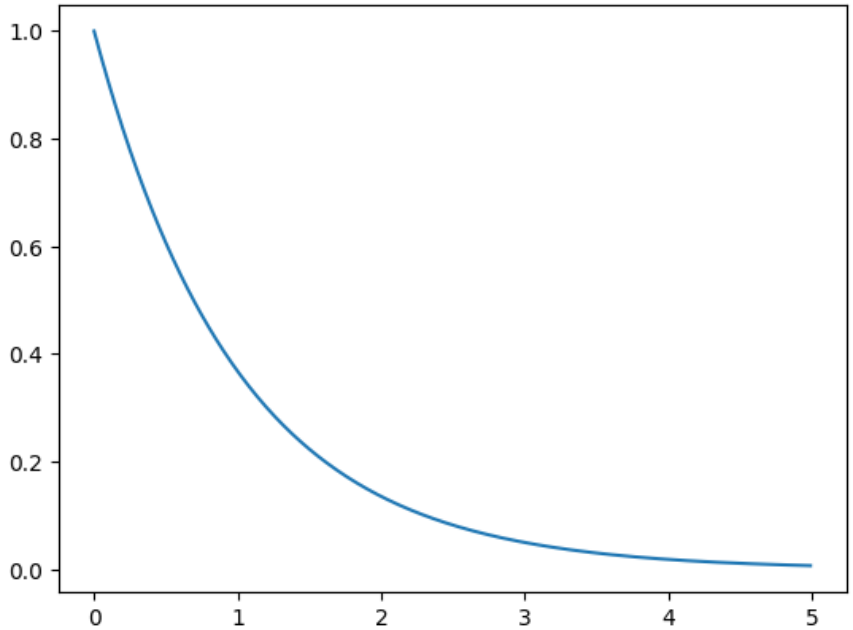
Note: In Python, Y (the log-normally distributed variable) is parameterized with $\mathrm{loc} = 0$, $s = \beta$ and $\mathrm{scale} = e^{\alpha}$

The Exponential Distribution

The exponential distribution is used to describe a continuous random variable X that follows an exponentially decreasing distribution and is often used to describe *lifetimes and waiting times*.

The exponential distribution can be used to describe the (waiting) time between events in a Poisson process.

$$X \sim Exp(\lambda)$$



Kahoot!

(x3)

The exponential distribution, Def. 2.48 & Thm. 2.49

Notation:

$X \sim \text{Exp}(\lambda)$, where $\lambda > 0$.

Density function:

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Mean:

$$\mu = \frac{1}{\lambda}$$

Variance:

$$\sigma^2 = \frac{1}{\lambda^2}$$

The exponential distribution

- The exponential distribution is often used to model lifetimes and waiting times.
- The exponential distribution can describe the (waiting) time between events in a Poisson process.
- The exponential distribution is a special case of the *gamma distribution*.

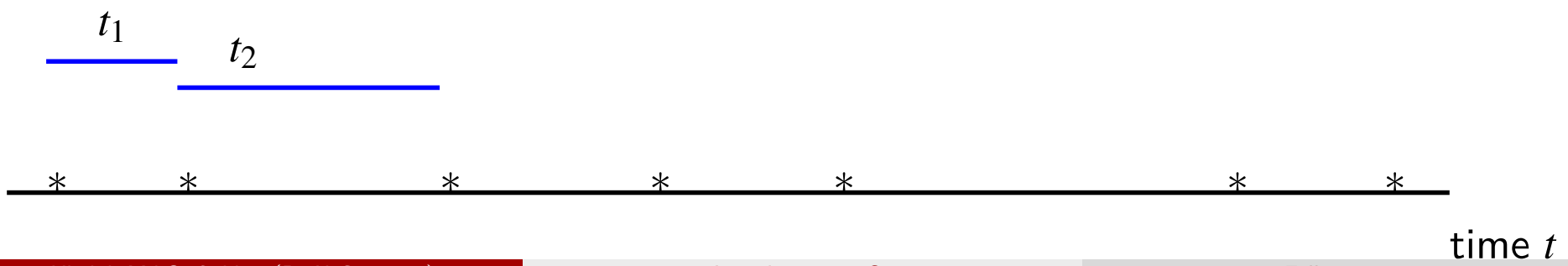
Connection between the exponential and Poisson distribution

The exponential is related to the Poisson distribution:

- The exponential distribution describes the time between random events, and this is a continuous quantity.
- The Poisson distribution describes the number of random events that occur within a fixed time period, and this is a discrete quantity.
- Some tasks can be solved with both the exponential distribution and the Poisson distribution.

Poisson: Number of events per (time) unit

Exponential: Continuous distance between events



Example 4

Queue model: Poisson process

The time between customer arrivals at a post office follows an exponential distribution with a mean of $\mu = 2$ minutes.

Question:

a) A customer has just arrived. What is the probability that no more customers will arrive within a period of 2 minutes?
*b) Now use the **Poisson** distribution to calculate the probability that no more customers will arrive within the next two minutes.*



Specific continuous distributions in Scipy.stats

```
Scipy.stats.uniform
Scipy.stats.normal
Scipy.stats.lognormal
Scipy.stats.expon
```

General 'methods' for different distributions are:

`scipy.stats`	`.uniform/.norm/.lognorm/.expon`
`.rvs`	'random variates' (simulate random numbers)
`.pdf`	'probability density function' (pdf/density function)
`.cdf`	'cumulative distribution function' (distribution function)
`.ppf`	'percent point function' (inverse cdf / quantile function)
`.mean /.var /.std`	'mean'/'variance'/'standard deviation'

See also Appendix A.2.1 in the book.

Example 5: Normal distribution in Python

Let's simulate the random variable X that follows a Normal distribution with mean 5 and standard deviation 2:

$$X \sim N(5, 2^2)$$

In Python we use Scipy.stats' `norm.rvs` function:

```
stats.norm.rvs(loc=5, scale=2)
```

- Go to the Python notebook "week3_python.ipynb" in VS Code



Visual Studio Code

Kahoot! (x1)

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Calculations Involving Random Variables

When performing calculations that involve a random variable, the result is a new random variable.

Examples:

Let X be a random variable.

If $Y = 2X + 10$, then Y is also a random variable.

If $Z = 3X^2$, then Z is also a random variable.

If $W = 5\cos(X)$, then W is also a random variable.

Rules for stochastic variables

These rules apply to both continuous and discrete random variables!

Let X be a random variable, while a and b are constants.

Mean rule:

$$\mathrm{E}[aX+b] = a\mathrm{E}[X]+b$$

Variance rule:

$$\mathrm{V}[aX+b] = a^2\mathrm{V}[X]$$

Example 6

Let X be a random variable with mean $\mathrm{E}[X] = 4$ and variance $\mathrm{V}[X] = 6$.

Question:

Calculate the mean and variance of $Y = -3X + 2$.



Example 6

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Question:

Calculate the mean and variance of $Y = -3X + 2$.

Answer:

$$\mathrm{E}[Y] = -3\mathrm{E}[X] + 2 = -3 \cdot 4 + 2 = -10$$

$$\mathrm{V}[Y] = (-3)^2 \mathrm{V}[X] = 9 \cdot 6 = 54$$



Rules for Random Variables

Let $X_1, \ldots, X_n$ be *independent* random variables.

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Mean Rule:

$$\begin{aligned} &\mathrm{E}[a_1 X_1 + a_2 X_2 + \cdots + a_n X_n] \\ &= a_1 \mathrm{E}[X_1] + a_2 \mathrm{E}[X_2] + \cdots + a_n \mathrm{E}[X_n] \end{aligned}$$

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Variance Rule:

$$\begin{aligned} &\mathrm{V}[a_1X_1 + a_2X_2 + \cdots + a_nX_n] \\ &= a_1^2\mathrm{V}[X_1] + \cdots + a_n^2\mathrm{V}[X_n] \end{aligned}$$

Example 7

Planning for Airline

The individual weight of passengers on a flight, X, is assumed to be normally distributed as $X \sim N(70, 10^2)$.

A plane that can take 55 passengers can be loaded with a maximum of 4000 kg (only the passengers' weight is considered here as the load).

Example 7

Planning for Airline

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Calculate the probability that the plane will be overloaded.

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Question:

Calculate the probability that the plane will be overloaded.

What is the total weight Y of 55 passengers on a flight?

How do we mathematically describe the random variable Y?

Example 7

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A plane that can take 55 passengers can be loaded with a maximum of 4000 kg (only the passengers' weight is considered here as the load).

Question:

Calculate the probability that the plane will be overloaded.

What is the total weight Y of 55 passengers on a flight?

How do we mathematically describe the random variable Y?

...

NOT: $Y = 55 \cdot X$

Example 7

What is the total weight Y of 55 passengers on a flight?

$Y = \sum_{i=1}^{55} X_i$, where $X_i \sim N(70, 10^2)$ (assumed to be independent)

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Example 7

What is the total weight Y of 55 passengers on a flight?

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Mean and Variance of Y:

$$\mathrm{E}[Y] = \sum_{i=1}^{55} \mathrm{E}[X_i] = \sum_{i=1}^{55} 70 = 55 \cdot 70 = 3850$$

$$\mathrm{V}[Y] = \sum_{i=1}^{55} \mathrm{V}[X_i] = \sum_{i=1}^{55} 100 = 55 \cdot 100 = 5500$$

Example 7

What is the total weight Y of 55 passengers on a flight?

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Mean and Variance of Y:

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$$\mathrm{V}[Y] = \sum_{i=1}^{55} \mathrm{V}[X_i] = \sum_{i=1}^{55} 100 = 55 \cdot 100 = 5500$$

Y is normally distributed, so we can find $P(Y > 4000)$ by:

$1 - F(y = 4000; \mu = 3850, \sigma^2 = 5500) = 0.02156$ (try it yourself in Python)

Example 7 - WRONG Analysis

What is Y?

NOT: $Y = 55 \cdot X$

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Mean and Variance of WRONG Y:

$$\mathrm{E}[Y] = 55 \cdot 70 = 3850$$

$$\mathrm{V}[Y] = 55^2 \mathrm{V}[X] = 55^2 \cdot 100 = 302500$$

Example 7 - WRONG Analysis

What is Y?

NOT: $Y = 55 \cdot X$

Mean and Variance of WRONG Y:

$$\mathrm{E}[Y] = 55 \cdot 70 = 3850$$

$$\mathrm{V}[Y] = 55^2 \mathrm{V}[X] = 55^2 \cdot 100 = 302500$$

The WRONG Y is also normally distributed. Here we find $P(Y > 4000)$ with WRONG Y:

$1 - F(y = 4000; \mu = 3850, \sigma^2 = 550^2) = 0.3925$ (try it yourself in Python)

Example 7 - WRONG Analysis

What is Y?

NOT: $Y = 55 \cdot X$

Mean and Variance of WRONG Y:

$$\mathrm{E}[Y] = 55 \cdot 70 = 3850$$

$$\mathrm{V}[Y] = 55^2 \mathrm{V}[X] = 55^2 \cdot 100 = 302500$$

The WRONG Y is also normally distributed. Here we find $P(Y > 4000)$ with WRONG Y:

$1 - F(y = 4000; \mu = 3850, \sigma^2 = 550^2) = 0.3925$ (try it yourself in Python)

Consequence of incorrect calculation:

MANY wasted money for the airline!!!

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χ^2-distribution

χ^2-distribution is used to describe a continuous random variable X, which is equal to the *sum of ν squared independent standard normally distributed variables*:

$$X \sim \chi^2(\nu) \qquad \text{where } X = \sum_{i=1}^{\nu} Z_i^2 \text{ and } Z_i \sim N(0, 1^2)$$

Theorem 2.79

Let $Z_1, \ldots, Z_\nu$ be independent random variables following the standard normal distribution, then

$$\sum_{i=1}^{\nu} Z_i^2 \sim \chi^2(\nu). \tag{2-94}$$

χ^2-distribution is only defined for non-negative values and is "right-skewed", but the shape only depends on the parameter ν

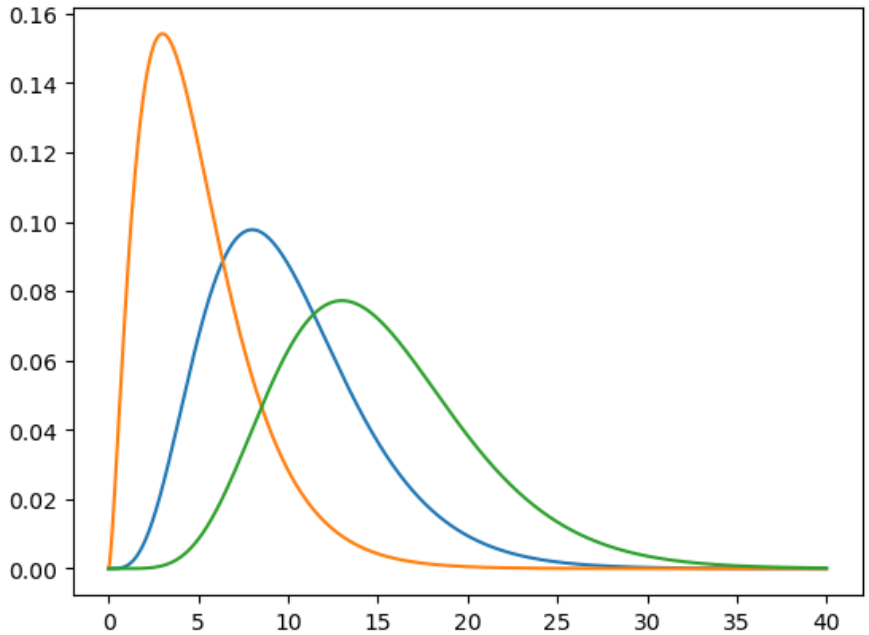
χ^2-distribution density function

Definition 2.78

Let X be χ^2 distributed, then its *pdf* is

$$f(x) = \frac{1}{2^{\frac{\nu}{2}}\Gamma\left(\frac{\nu}{2}\right)} x^{\frac{\nu}{2}-1} e^{-\frac{x}{2}}; \quad x \geq 0, \tag{2-93}$$

where $\Gamma\left(\frac{\nu}{2}\right)$ is the Γ-function and ν is the degrees of freedom.



Kahoot!

(x1)

t-distribution

The t-distribution is used to describe a continuous random variable X that is equal to *a standard normally distributed variable (Z) divided by square root of an independent χ^2-distributed variable (Y) scaled by the parameter ν*:

$$X \sim t(\nu)$$

Theorem 2.87

Let $Z \sim N(0,1)$ and $Y \sim \chi^2(\nu)$, then

$$X = \frac{Z}{\sqrt{Y/\nu}} \sim t(\nu). \tag{2-99}$$

The t-distribution is symmetric, similar to the standard normal distribution (especially if the parameter ν is very large), but the t-distribution has slightly *thicker tails*.

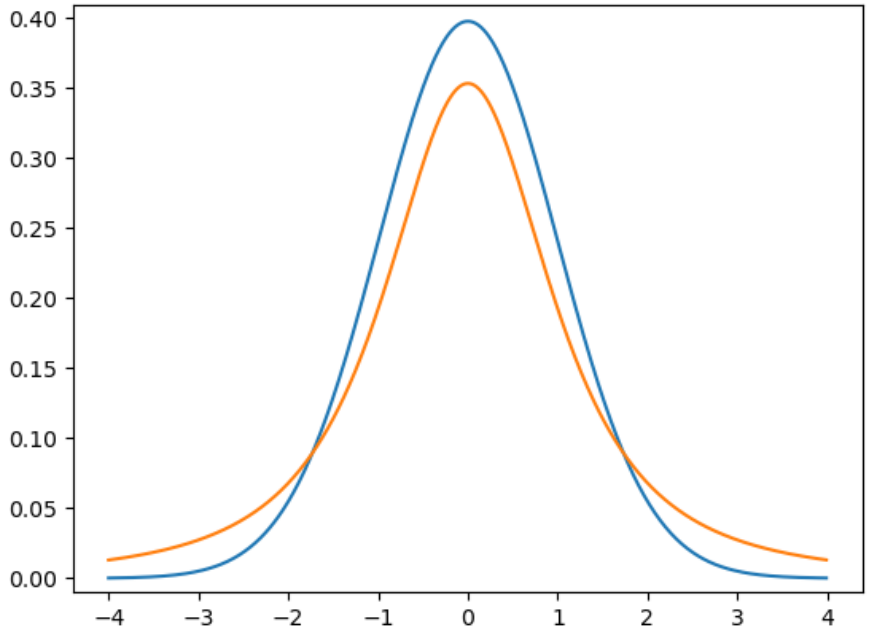
t-distribution density function

Definition 2.86

The t-distribution *pdf* is

$$f_T(t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)}\left(1+\frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}, \tag{2-98}$$

where ν is the degrees of freedom and $\Gamma()$ is the Gamma function.



Kahoot!

(x1)

F-distribution

F-distribution is used to describe a continuous random variable X that is equal to *the ratio of two independent χ^2-distributed variables (each scaled by their parameter ν)*:

$$X \sim F(\nu_1, \nu_2)$$

Theorem 2.96

Let $U \sim \chi^2(\nu_1)$ and $V \sim \chi^2(\nu_2)$, be independent then

$$F = \frac{U/\nu_1}{V/\nu_2} \sim F(\nu_1, \nu_2). \tag{2-105}$$

The F-distribution is only defined for non-negative values and is "right-skewed", but the shape depends on the parameters ν_1 and ν_2.

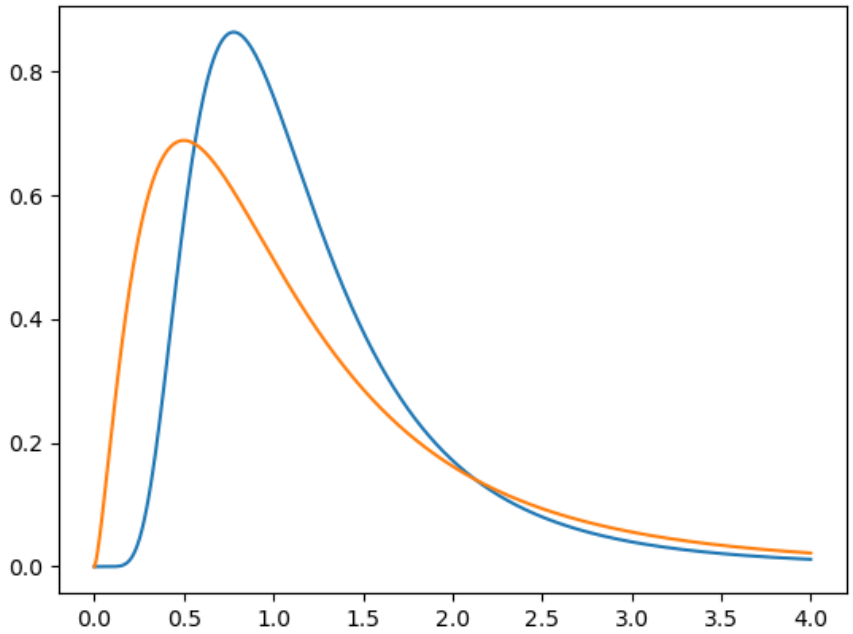
F-distribution density function

Definition 2.95

The F-distribution *pdf* is

$$f_F(x) = \frac{1}{B\left(\frac{\nu_1}{2}, \frac{\nu_2}{2}\right)} \left(\frac{\nu_1}{\nu_2}\right)^{\frac{\nu_1}{2}} x^{\frac{\nu_1}{2}-1} \left(1 + \frac{\nu_1}{\nu_2} x\right)^{-\frac{\nu_1+\nu_2}{2}}, \qquad (2\text{-}104)$$

where ν_1 an ν_2 are the degrees of freedom and $B(\cdot, \cdot)$ is the Beta function.



Special distributions in Scipy.stats

```
Scipy.stats.chi2
Scipy.stats.t
Scipy.stats.f
```

General 'methods' for different distributions are:

scipy.stats	.uniform/.norm/.lognorm/.expon
.rvs	'random variates' (simulate random numbers)
.pdf	'probability density function' (pdf/density function)
.cdf	'cumulative distribution function' (distribution function)
.ppf	'percent point function' (inverse cdf / quantile function)
.mean /.var /.std	'mean'/'variance'/'standard deviation'

See also Appendix A.2.1 in the book.

Kahoot!

(x1)

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